Brain Tumor Detection Using Transfer Learning

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***Abstract*—Brain tumor detection plays a crucial role in medical diagnosis, as early identification significantly improves treatment planning and patient survival rates. Brain tumors are caused by abnormal growth of cells within brain tissues and may be classified as benign or malignant. Magnetic Resonance Imaging (MRI) is the most commonly used imaging technique for brain tumor diagnosis due to its superior soft tissue visualization and non-invasive nature. However, manual interpretation of MRI scans is time-consuming, depends heavily on radiologist expertise, and is susceptible to diagnostic variability.**

Recent advances in deep learning have enabled automated image classification systems with high performance. Convolutional Neural Networks (CNNs) are particularly effective for extracting hierarchical image features. However, training CNNs from scratch requires large annotated datasets and significant computational resources. Transfer learning addresses this limitation by utilizing pretrained models trained on large-scale datasets.

In this work, a transfer learning-based framework using the VGG16 architecture is proposed for brain tumor classification. The model utilizes ImageNet-pretrained weights and fine-tunes higher convolutional layers for domain-specific learning. A dataset of 7200 MRI brain images categorized into four classes (glioma, meningioma, pituitary tumor, and no tumor) is used for experimental evaluation.

Experimental results demonstrate a classification accuracy of 95%, with strong precision, recall, and F1-score values. Confusion matrix and ROC-AUC analysis further validate the effectiveness of the proposed approach.

***Index Terms*—Brain Tumor Detection, MRI, Deep Learning, CNN, Transfer Learning, VGG16**

I Introduction

Brain tumors are among the most life-threatening neurological disorders affecting millions of people worldwide. A brain tumor refers to an uncontrolled growth of abnormal cells inside the brain, which can disrupt normal brain functioning and lead to severe complications. Depending on their growth behavior, brain tumors are categorized into benign and malignant tumors. Benign tumors are non-cancerous and grow slowly, whereas malignant tumors are cancerous, aggressive, and capable of spreading to surrounding tissues.

Common symptoms associated with brain tumors include headaches, seizures, vision impairment, vomiting, memory loss, and muscle weakness. Early diagnosis is critical because delayed detection can significantly reduce treatment effectiveness.

Magnetic Resonance Imaging (MRI) is considered the gold standard for brain tumor diagnosis due to its ability to produce high-resolution images of soft tissues. MRI scans provide detailed information about tumor size, shape, and location. However, manual analysis of MRI scans by radiologists is labor-intensive and prone to inter-observer variability.

Artificial Intelligence (AI), particularly deep learning, has emerged as a promising solution for automated medical diagnosis. CNNs have achieved remarkable performance in image classification tasks due to their ability to automatically learn spatial hierarchies of features. Transfer learning further improves performance by leveraging pretrained models.

This research proposes a transfer learning-based VGG16 model for automated brain tumor classification.

II Related Work

Deep learning has transformed medical image analysis by enabling automated feature extraction and classification. CNN architectures such as AlexNet, ResNet, DenseNet, and VGGNet have been widely adopted in medical applications.

LeCun et al. demonstrated the effectiveness of deep learning in image recognition tasks. ResNet introduced residual learning, enabling deeper networks and improved performance. U-Net became highly successful for biomedical image segmentation due to its encoder-decoder structure.

Transfer learning has shown significant success in medical imaging, especially when dataset size is limited. By reusing features learned from large datasets such as ImageNet, models can generalize better with fewer training samples.

In brain tumor classification, previous studies have reported accuracy ranging from 85% to 98%, depending on architecture complexity, preprocessing techniques, and dataset quality.

Despite newer architectures, VGG16 remains highly effective because of its stable architecture and strong feature extraction capabilities.

III Methodology

III-A Dataset Description

The dataset used in this study consists of 7200 MRI brain images categorized into four classes:

- Glioma Tumor
- Meningioma Tumor
- Pituitary Tumor
- No Tumor

The dataset is divided into 5600 training images and 1600 testing images. The balanced distribution across classes improves model generalization.

All images are resized to 224×224 pixels to match VGG16 input dimensions.

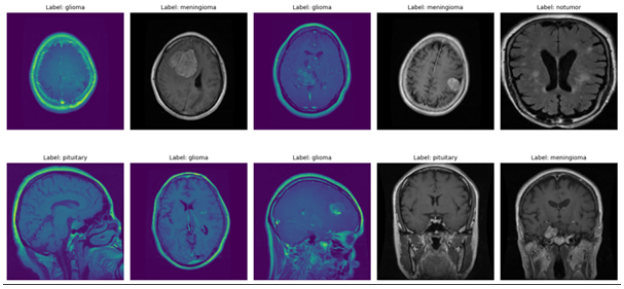

Fig. 1: Sample MRI Images from Dataset

III-B Data Preprocessing

Data preprocessing is essential for improving model stability and performance.

The following preprocessing steps are applied:

1. Image resizing to 224×224.
2. Pixel normalization to scale values between 0 and 1.
3. Dataset shuffling for randomness.
4. Batch creation for efficient training.

Normalization reduces computational instability and improves convergence speed.

IV Model Architecture

The proposed system uses the VGG16 architecture as the base feature extractor. VGG16 consists of 13 convolutional layers and 3 fully connected layers.

The architecture uses 3×3 convolution filters with ReLU activation and max-pooling layers.

The pretrained ImageNet weights help extract generic visual features such as edges and textures.

To adapt the model for brain tumor classification:

- Lower layers are frozen
- Final convolution layers are fine-tuned
- Custom dense layers are added

The classification head consists of:

- Global Average Pooling
- Dense Layer (128 neurons)
- Dropout (0.4)
- Softmax Output Layer

V Training Configuration

The model training uses the Adam optimizer due to its adaptive learning capability.

Training parameters:

- Epochs = 15
- Batch Size = 32
- Learning Rate = 1×10^{-4}
- Loss Function = Sparse Categorical Crossentropy

The model is trained using GPU acceleration to reduce computation time.

Fig. 2: Proposed VGG16-Based Architecture

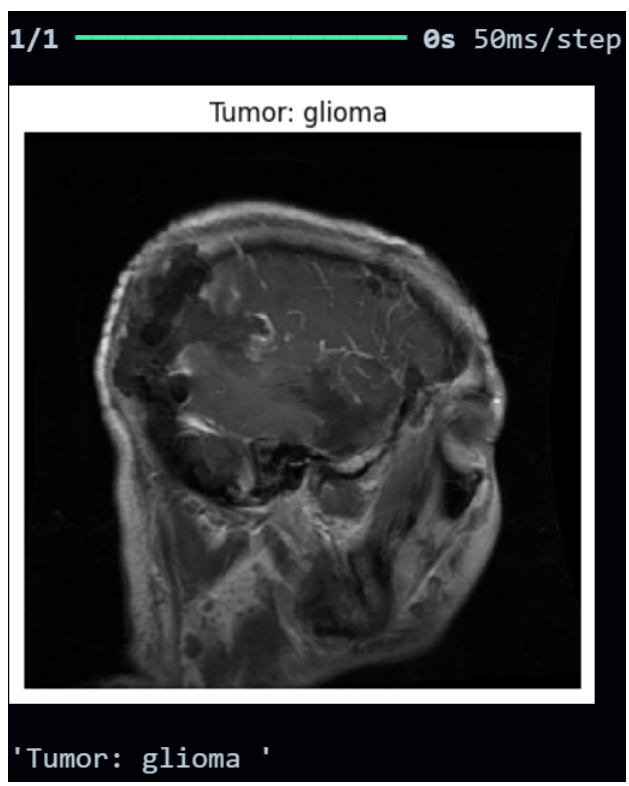

Fig. 3: Proposed VGG16-Based Architecture

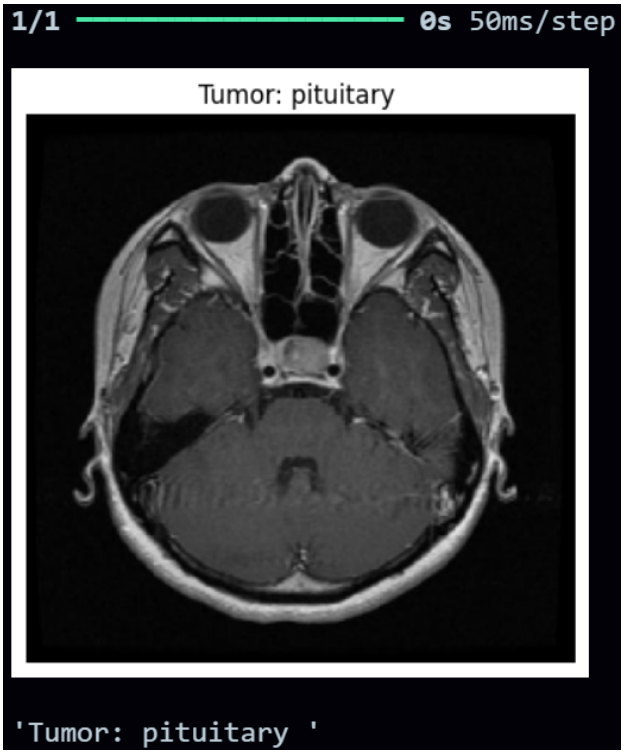

Fig. 4: Proposed VGG16-Based Architecture

VI Experimental Results

The proposed model demonstrated excellent training and testing performance.

Training accuracy reached 99.8%, while test accuracy reached 95%.

This indicates strong feature learning and good generalization.

VI-A Training Performance

The training accuracy increased consistently while loss decreased rapidly, showing effective optimization.

Fig. 5: Training Accuracy and Loss

VI-B Confusion Matrix Analysis

The confusion matrix shows strong classification performance across all categories.

Misclassifications were minimal, indicating robust feature extraction.

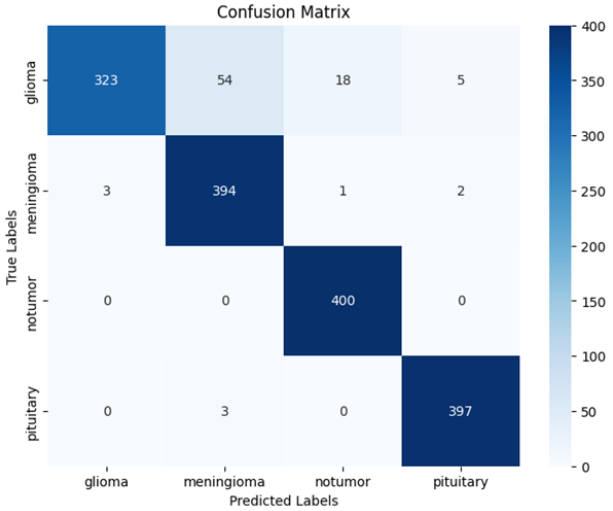

Fig. 6: Confusion Matrix

VI-C ROC-AUC Analysis

ROC curves evaluate class separability.

The AUC score for all classes exceeded 0.97, demonstrating excellent discriminative capability.

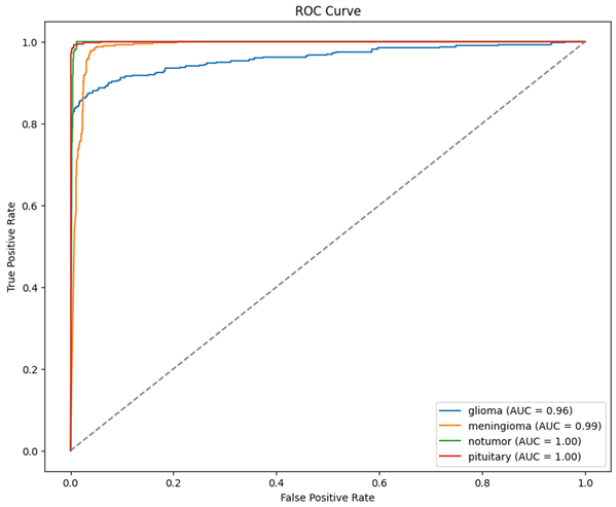

Fig. 7: ROC Curve Analysis

VII Discussion

The results confirm that transfer learning with VGG16 is highly effective for brain tumor classification. The pretrained layers accelerate convergence and improve generalization.

The model performed strongly across all tumor classes, especially in distinguishing tumor and non-tumor cases.

However, some limitations remain. The dataset size is moderate and may not fully represent real-world clinical diversity. Additionally, model interpretability remains a challenge.

Future integration of explainability techniques such as Grad-CAM can improve trustworthiness.

VIII Conclusion

This paper presented an automated brain tumor classification framework using transfer learning with VGG16.

The proposed model achieved 95% classification accuracy, demonstrating strong performance for MRI-based diagnosis.

The approach provides a computationally efficient and reliable solution for assisting radiologists.

Future work will focus on larger datasets, explainable AI techniques, and advanced architectures such as ResNet and Vision Transformers.

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